

A Packing of 7 Four-Spaces in $\mathbb{R}^{12}$

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Abstract

We give 7 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 3.110922. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 3.047366, obtained by deleting 1 of the 8 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 7 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 3.110922\dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 7$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 3.047366, obtained by deleting 1 of the 8 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 7/6 = 3.111111$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N7.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
259243532778162665877805499185993385290813673204288630299228958056651628
493924899945612263979163729782467350545212513389091513149154107646265632
116808495883211787288517314142232292596266078927
```

denominator:

```
833333333333172688490513616930983570554953691016057493066647383766504040
064662063566894837667150985182315060104601651409715188969428660458014653
13667093122673565419113891874025856785875838297
```

B Coordinates

7 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N7.txt`.

```
403547526091 -115474465836 -10963511464 -295150726477460083667 -152295513355 478133342567 43090930889124151483773 24253424171 -238369405713 -256230651934
-183870411546 -113995632591 -252996076111 -1932297388803498564 336781669377 -146652854261 15975518120357720243834 144546904962 181890550396 39227076162
359650195223 -31139423457 -412918112078 41586924533461257642639 310538440028 17985831091 6972556276 -541345551886 -56962672311 -125628060884 -161547333558
126379458151 -99490262942 450315416361 63786231133 -96954596345 -20258872965 -186255366851 -54514251064394320259159 559350885588 -157529476767 307634629420
105971387803 97879298270 -415978077636 -294544111869226157533165 378347753394 -214258197655 39105855184994727567243 -135529971824 -92296444253 -440418803957
-145377647718 273117456270 423181681135 -2930406397340003466828 -423429440782 -615529418111 36502184672035066584755 -778433901449 -354345278856 274308590829
-342812671410 -339215156482 -6254417679 -353299879320576676605 62991791579 349143209925 -78868048392 -7006672635 -221405313243 611780502312 261553345943
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-539084094492 -260959642444 29529231040 318853923756305919240270 43684116325 388293008943 -85827758095-18216132825 -145237565707 143447702876 -289948182769
-174037428720 313471306633 56965724375 532613230488-353068256115 283522940545 -549265808453 -26203270588071492852 -75504709221 -238200165926 -190915921584
100143645563 323520088531 242860243002 609943590711-255694466104 52821245551 188503723551 -24907874613295465802946 -334777164174 459035797534 -1413639712
199347313707 -303294291046 20259710144 72553802490 174824044681 601044088851 -21605383819 -514454965763684052806 -80015779581 -244176153181 -378448396735
210001336466 480021143098 482013609895 7907260206 -424042016544 259622133957 -368435008562 471450308638117939170 -276007653695 178934325859 -555254476738
300897699698 -291972469729 125791313536 3236809365215085422107 -100565091853 -311624975626 -947872150520163709245 -90659554287 9200613204 -32633904021
-360456841978 228444465406 26434582930 -883130624375937519465 -140766950689 -215193518240 235889133229470837591219 113781373993 513114153840 -583612488031
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-391672563261 -318862247230 520451825507 6868910482364945889943 -306008036116 -210518224225 -4244562231886140175950 -55174489038 -275212796893 -252850061797
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359682469264 112034073852 -147888919730 266755845632617727406 62105885001 -225594418329 232397107847 663698347266 26526197375 247096145319 247898669367
53844911926 21457258812 77297274589 285411424700 -124096067130 -176148619421 730167769545 16305649524223674183836 -298844569852 85174771353 320580045872
275658720331 64726277845 -321625418971 -10497385681850830550334 -131573544864 121045344533 21578924326754837996447 37936840143 -177426616396 207523803517
224060452745 -533323725525 285254959057 -22960210330717657567 63833066084 217142603637 33807550756 235416719728 -55055320239 33378028252 -72853317682
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558130368344 114951490125 4069453103 59715842355 -142380017314 -482407410230 -283041229410 4144602376491794644 -176952651835 282208215910 -2017142126687
26120754688 -470753899776 220256083105 26984055277401510730838 134395479063 -238018316088 547143673977504072582 603323089762 289753384810 3162919735
```